

Parthapratim Chatterjee

Third Year Undergraduate, IIT Kanpur
Department of Material Science and Engineering

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Academic Qualifications

Year	Educational Qualification	Institute	CPI/%
2021 - Present	B.Tech	Indian Institute of Technology Kanpur	7.60/10
2021	CBSE(XII)	Mansukhbhai Kothari National School, Pune	95.2%
2019	ICSE(X)	Vibgyor High International School NIBM, Pune	95.4%

Scholastic Achievements

- Secured **All India Rank 5980** in **Joint Entrance Examination Advanced 2021** among the 1.5 Lakh shortlisted candidates.
- Secured **All India Rank 6858** in **Joint Entrance Examination Mains 2021** among the 1 Million shortlisted candidates.

Work Experience

Generation of 3D objects from 2D image | Internship | *Motherson Technology Services Limited, Noida* December'23

Objective	• To generate a 3D CAD compatible design from 2D sketch and given desired design parameters.
Approach	• Designed a Generative AI network to train using available design datasets to generate 3D objects from 2D images. • AI-generated 3D objects are compatible with CAD design software for further editing and accuracy. improvement.
Result	• Generated a AI engine that could generate 3D design from 2D input which is compatible with CAD softwares.

Synthetic Microstructure Data Generation For Machine Learning Composite Architecture Prediction | **SURGE'23**
Prof. Shikhar Krishn Jha May'23-July'23

Objective	• To predict properties of materials computationally as an alternative to physical experimentation using dataset.
Approach	• Designed an algorithm in Python to generate Synthetic Microstructure images to form a dataset for the input. • Automated the FEniCS code to generate the Material Property plots for the dataset of microstructure images. • Trained GAN model for the prediction of property plots from input image using Image to Image translation.
Result	• Generated a diverse dataset of images of Microstructures for input to GAN model for training and prediction. • Increasing iterations led to highly accurate model-generated material plots, confirmed by minimal RMS error .

Key Projects

- Planning and Execution of Grasping Motion** | *Team Humanoid, IIT Kanpur* | *Prof. K S Venkatesh* January'23-April'23
 - Conducted **torque analysis** to estimate load on bot's arm, calculated required **current** and **voltage** for gripper's holding capacity.
 - Applied an effective **Stereo-Block Matching algorithm** and **YOLOv5** for the detection of objects, their depth, and dimensions.
 - Implemented **MiDaS** for **monocular depth estimation** using a single camera setup which provides a **depth map** for images.
- Computer Vision and Sensors of AUTOMI 🤖** | *Team Humanoid, IIT Kanpur* | *Prof. K S Venkatesh* December'22
 - Trained a custom **YOLOv5** model to accurately detect **Ping-Pong balls** using a custom dataset with **augmented images**.
 - Implemented method to **detect center** of circle of revolving archery board and **trajectory** that was followed was stored.
 - Developed code to **track lines** and **change direction** based on detected arrows, such as turning left when detecting a left arrow.
- Brain Tumor Detection 🧠** | *Self Project* December'22-January'23
 - This Brain Tumor Detection application employs **Convolutional Neural Networks** to identify brain tumors from MRI images.
 - Learned about **neural networks** and concepts of **CNN** including **activation functions**, **strides**, **padding** and **convolutions**.
 - Achieved a remarkable **99 %** testing accuracy by implementing a **VGG16 model** on Brain MRI images for detection of tumor.
- Regression Analysis 📊** | *Stamatics Society, IIT Kanpur* May'22-July'22
 - Implemented ML algorithms such as **Linear**, **Logistic** and **Polynomial Regression** with **Cost function** and **Gradient Descent**.
 - Implemented a **Logistic Regression** model from scratch to predict diabetes from a dataset using BMI, Age and other factors.
 - Constructed a **Multivariate Regression model** with **R2 score of 0.76** to **accurately predict** Insurance price from **5** factors.
 - Demonstrated rising CO2 levels by analyzing and fitting emission data with **polynomial curves of various degrees** in Python.

Technical Skills

Programming Languages	Operating Systems and Softwares	Tools and Libraries
C, C++, Python, Java, MySql, MATLAB, L ^A T _E X, HTML	Windows, Ubuntu, Autodesk Fusion 360, MacOS, Microsoft Office	Numpy, Pandas, Matplotlib, Tensorflow, Keras, OpenCV, SciPy, Git, Scikit-Learn

Positions of Responsibility

- DUGC Students' Nominee, Materials Science and Engineering** | **Students Gymkhana, IIT Kanpur** Sept'23- Present
 - Collaborated with **professors** of the department for handling any problems faced by students in their academic programmes.
- Manager, Research and Projects** | **Materials Society, IIT Kanpur** Jul'23- Present
 - Organised various **Academic Oriented** events like **Alumni talks**, **Fresher Orientation**, **Farewell** and **Intern Sessions**.
- Team Member** | **Team Humanoid, IIT Kanpur** Aug'22-May'23
 - Helped to organise events like **SnT Pavilion** where the team's past work was shown in front of **1000+** students over three days.

Relevant Coursework

Fundamentals of Computing (A)	Introduction to Electronics	Partial Differential Equations
Linear Algebra and Ordinary Differential Equations	Supervised Machine Learning Regression and Classification*	Python for Computer Vision with OpenCV and Deep Learning*
Computational Methods in Engineering	Data Structures and Algorithms	Deep Learning Onramp*

Extra-Curricular Activities

- Secured **First** position in **Inter House Cricket Competition** and **First** position in **Presentation of a Speech** in School.
- Participated in **3rd Pauling Memorial Inter-School Science Quiz** organized by **IISER, Pune** on **National Science Day**.
- Coordinated in conducting **Technocruise** which is a pre zonal round of **Techkriti** as a **Senior Executive** in **Techkriti**.
- Completed the **MUNtor Program** and received the beginner certification from **MUN Cafe** for excelling in debate and diplomacy.